

REPLIT PROMPT — YachtWorth: Yacht Verification Module

CONTEXT

Build a new screen "Yacht Verification" inside the existing YachtWorth app. Entry point: Tools tab → Yacht Verification card (currently SOON → change to LIVE after build). This module verifies yacht data against external maritime registries and databases. Data source: My Yacht profile (primary) OR manual entry.

OVERVIEW

Three verification levels:

Level 1 — Basic Auto (Free plan) IMO number lookup + flag state basic check.

Level 2 — Full Auto (Pro plan) IMO + AIS history + flag registry + ownership history.

Level 3 — Professional (Coming Soon) All auto checks + human surveyor assigned via PDYE platform.

EXTERNAL APIs

1. VesselFinder API — AIS data + vessel info

- Endpoint: `https://api.vesselfinder.com/vessels`
- Used for: current position, AIS history, vessel details
- Auth: API key in header
- Cost: €100/month (Professional plan, 5,000 calls)
- Docs: <https://api.vesselfinder.com/docs>

2. IMO Ship Lookup — vessel registration

- Use VesselFinder vessel details endpoint (includes IMO data)

- Fallback: public IMO GISIS database scrape for basic lookup

3. Equasis API — flag state + safety records (FREE)

- Endpoint: <https://www.equasis.org/EquasisWeb/restricted/Search>
- Requires free registration at [equasis.org](https://www.equasis.org)
- Returns: flag state, classification society, PSC inspections, deficiencies
- Auth: username + password in POST body

4. MarineTraffic API (optional, secondary)

- Used only if VesselFinder returns no results
- Endpoint: <https://services.marinetraffic.com/api>
- Auth: API key

SUPABASE SCHEMA

```
create table if not exists yacht_verifications (  
  id uuid primary key default gen_random_uuid(),  
  yacht_id uuid references yachts(id) on delete cascade,  
  user_id text not null,  
  
  -- Input data used for verification  
  imo_number text,  
  vessel_name text,  
  flag text,  
  length_m numeric,  
  year_built integer,  
  
  -- Verification results  
  verification_level integer default 1, -- 1, 2, or 3  
  overall_status text default 'unverified', -- unverified | basic | verified | fa  
  
  -- IMO check  
  imo_valid boolean,  
  imo_matches_name boolean,  
  imo_matches_flag boolean,  
  imo_registered_owner text,  
  imo_gross_tonnage numeric,  
  imo_vessel_type text,
```

```

imo_flag_history jsonb, -- array of past flags

-- AIS check
ais_found boolean,
ais_last_position jsonb, -- {lat, lng, timestamp, port}
ais_current_status text, -- 'moored' | 'underway' | 'at anchor' | 'unknown'
ais_last_port text,
ais_speed numeric,
ais_history_months integer, -- how many months of history checked

-- Flag registry check
flag_registry_found boolean,
flag_registry_status text, -- 'active' | 'suspended' | 'deleted' | 'unknown'
flag_registry_valid_until date,
flag_class_society text,

-- Equasis safety check
equasis_found boolean,
equasis_deficiencies integer default 0,
equasis_detentions integer default 0,
equasis_last_inspection date,
equasis_inspection_result text,

-- Ownership / lien check
lien_check_performed boolean default false,
lien_found boolean,
lien_details text,

-- Overall score
verification_score integer, -- 0-100
verification_badges jsonb, -- array of earned badges

-- Meta
verified_at timestampz default now(),
expires_at timestampz, -- 90 days validity
raw_response jsonb -- full API responses for audit
);

create index verif_yacht_id_idx on yacht_verifications(yacht_id);
create index verif_user_id_idx on yacht_verifications(user_id);

```

NAVIGATION

Tools tab

- Yacht Verification (YachtVerificationScreen.js)
 - VerificationInputScreen (if no My Yacht selected)
 - VerificationRunningScreen (progress)
 - VerificationResultScreen (results + badges)
 - VerificationDetailScreen (per-check details)
-

SCREEN 1: ENTRY

YACHT VERIFICATION

IMO · AIS · Flag Registry

[Select from My Yacht ▼]

Aurelia · Azimut 24m · IMO 9812345

— or —

[Enter manually]

VERIFICATION LEVEL:

★ Basic	FREE
IMO lookup · Flag check	
★★★ Full	PRO
IMO · AIS history · Safety records	

[Start Verification →]

SCREEN 2: MANUAL INPUT (if no yacht selected)

VESSEL DETAILS

IMO NUMBER *

[_____]

 7-digit number on vessel documents

VESSEL NAME

[_____]

FLAG / COUNTRY

[_____]

LENGTH (m)

[_____]

YEAR BUILT

[_____]

SCREEN 3: VERIFICATION RUNNING

Verifying vessel...

Checking IMO registry... ✓ 2.1s

Checking AIS data... ✓ 3.4s

Checking flag registry... ✓ 1.8s

Checking safety records... ⌚

Analysing results...

This usually takes 15-30 seconds

Progress shown as checks complete in real time. Each check updates independently as API responds.

SCREEN 4: VERIFICATION RESULT

Header — overall status:

Verified (score 80-100):

✓ VERIFIED
Verification Score: 94/100
Checked: 28 May 2026
Valid until: 25 Aug 2026 (90 days)

Partially verified (score 40-79):

[_____]

● PARTIALLY VERIFIED
Score: 62/100
2 checks passed · 1 warning

Failed / issues found:

△ ATTENTION REQUIRED
Score: 31/100
Issues found — review details

Verification Badges (earned on pass):

VERIFICATION BADGES

[✓ IMO Registered]	[✓ Flag Active]
[✓ AIS Active]	[✓ No Detentions]
[✓ Position Confirmed]	

Badges are grey/locked if check not passed. Gold border on earned badges.

Check Results — cards:

IMO Check card:

IMO REGISTRATION ✓ PASS

IMO Number	9812345
Registered name	Aurelia
Registered flag	Malta ✓ matches
Gross tonnage	98 GT
Vessel type	Motor Yacht ✓ matches
Registered owner	[shown if available]
Flag history	Malta (2019 – present)

[View full IMO record →]

AIS Check card:

AIS TRACKING ✓ ACTIVE

Last seen	2 hours ago
Current status	Moored
Last port	Port Vieux, Antibes
Coordinates	43.5805° N, 7.1239° E
Speed	0.0 kn
AIS history	12 months available

[View on map →]

Flag Registry card:

FLAG REGISTRY ✓ VALID

Flag state	Malta
Registry status	Active
Class society	Bureau Veritas
Valid until	Dec 2026

Safety Records card:

SAFETY RECORDS (Equasis) ✓ CLEAN

Deficiencies	0
Detentions	0
Last PSC inspection	Mar 2025
Result	No deficiencies found

Lien Check card (Pro only):

MARITIME LIENS ✓ CLEAR

Jurisdiction	Malta Maritime Authority
Mortgage/liens	None found
Arrest history	None found

⚠ Note: this check covers Malta registry only.
Verify with lawyer for comprehensive lien search.

Actions at bottom:

[ Download Certificate]

[ Add to Yacht Passport]

[ Re-verify]

[Share report]

VERIFICATION CERTIFICATE PDF

Generate a clean one-page PDF certificate:

YACHTWORTH

VERIFICATION CERTIFICATE

VERIFIED ✓

Score: 94/100

Vessel: Aurelia

IMO: 9812345

Flag: Malta

Type: Motor Yacht 24m

Checks performed:

- ✓ IMO Registration – Valid
- ✓ AIS Tracking – Active (Antibes, 2h ago)
- ✓ Flag Registry – Active, BV class
- ✓ Safety Records – No deficiencies
- ✓ Maritime Liens – Clear (Malta)

Verified by: YachtWorth AI Verification

Date: 28 May 2026

Valid until: 25 Aug 2026

Certificate ID: YW-2026-94821

This certificate is generated automatically from public maritime databases. Not a substitute for professional survey or legal title search.

YACHTWORTH · Powered by PDYE Group

API INTEGRATION CODE

```
// /utils/verificationAPI.js
```



```

const VESSELFINDER_KEY = process.env.VESSELFINDER_API_KEY;
const EQUASIS_USER = process.env.EQUASIS_USERNAME;
const EQUASIS_PASS = process.env.EQUASIS_PASSWORD;

// — IMO + AIS via VesselFinder —
export async function checkVesselFinder(imoNumber, vesselName) {
  try {
    // Search by IMO
    const response = await fetch(
      `https://api.vesselfinder.com/vessels?userkey=${VESSELFINDER_KEY}&imo=${im
        { headers: { 'Accept': 'application/json' } }
    );
    const data = await response.json();

    if (!data || data.length === 0) {
      // Try by name if IMO not found
      const byName = await fetch(
        `https://api.vesselfinder.com/vessels?userkey=${VESSELFINDER_KEY}&name=$
      );
      const nameData = await byName.json();
      return processVFResponse(nameData[0] || null);
    }

    return processVFResponse(data[0]);
  } catch (error) {
    return { found: false, error: error.message };
  }
}

function processVFResponse(vessel) {
  if (!vessel) return { found: false };
  return {
    found: true,
    imo: vessel.IMO,
    name: vessel.NAME,
    flag: vessel.FLAG,
    type: vessel.TYPE_NAME,
    grossTonnage: vessel.GT,
    lat: vessel.LAT,
    lng: vessel.LON,
    speed: vessel.SPEED,
    status: vessel.STATUS,
    lastPort: vessel.LAST_PORT,
    lastSeen: vessel.TIMESTAMP,
    aisActive: vessel.TIMESTAMP
      ? (Date.now() - new Date(vessel.TIMESTAMP).getTime()) < 7 * 24 * 60 * 60 *

```

```

        : false,
    };
}

// — Equasis safety check —
export async function checkEquasis(imoNumber) {
    try {
        const formData = new URLSearchParams({
            'P_IMO': imoNumber,
            'P_SEARCH_TYPE': 'VESSEL',
            'submit': 'Search',
        });

        const response = await fetch(
            'https://www.equasis.org/EquasisWeb/restricted/ShipInfo',
            {
                method: 'POST',
                headers: {
                    'Content-Type': 'application/x-www-form-urlencoded',
                    'Authorization': 'Basic ' + btoa(`${EQUASIS_USER}:${EQUASIS_PASS}`),
                },
                body: formData.toString(),
            }
        );

        const html = await response.text();
        return parseEquasisHTML(html);
    } catch (error) {
        return { found: false, error: error.message };
    }
}

function parseEquasisHTML(html) {
    // Parse key fields from Equasis HTML response
    // Look for: deficiencies count, detention count, last inspection date
    const defMatch = html.match(/Deficiencies.*?(\\d+)/s);
    const detMatch = html.match(/Detentions.*?(\\d+)/s);

    return {
        found: !html.includes('No vessel found'),
        deficiencies: defMatch ? parseInt(defMatch[1]) : 0,
        detentions: detMatch ? parseInt(detMatch[1]) : 0,
    };
}

// — Calculate verification score —
export function calculateScore(results) {

```

```

let score = 0;
let maxScore = 0;

const checks = [
  { key: 'imoFound',      weight: 25, value: results.imo?.found },
  { key: 'imoMatchesName', weight: 15, value: results.imo?.name?.toLowerCase() },
  { key: 'imoMatchesFlag', weight: 10, value: results.imo?.flag === results.inp },
  { key: 'aisFound',      weight: 20, value: results.ais?.found },
  { key: 'aisActive',     weight: 10, value: results.ais?.aisActive },
  { key: 'equasisClean',  weight: 20, value: results.safety?.found && results.
];

for (const check of checks) {
  maxScore += check.weight;
  if (check.value) score += check.weight;
}

return Math.round((score / maxScore) * 100);
}

// — Main verification function —
export async function verifyYacht(yacht, level = 1) {
  const results = {
    inputName: yacht.name,
    inputIMO: yacht.imo_number,
    inputFlag: yacht.flag,
    imo: null,
    ais: null,
    safety: null,
  };

  // Run checks in parallel
  const [vfResult, equasisResult] = await Promise.allSettled([
    checkVesselFinder(yacht.imo_number, yacht.name),
    level >= 2 ? checkEquasis(yacht.imo_number) : Promise.resolve(null),
  ]);

  results.imo = vfResult.status === 'fulfilled' ? vfResult.value : { found: false };
  results.ais = vfResult.status === 'fulfilled' ? vfResult.value : { found: false };
  results.safety = equasisResult.status === 'fulfilled' ? equasisResult.value : null;

  results.score = calculateScore(results);
  results.status = results.score >= 80 ? 'verified'
    : results.score >= 40 ? 'partial'
    : 'failed';

  return results;
}

```

```
}
```

SAVE TO SUPABASE

```
export async function saveVerification(yachtId, userId, results) {
  const expiresAt = new Date();
  expiresAt.setDate(expiresAt.getDate() + 90); // valid 90 days

  const { data, error } = await supabase
    .from('yacht_verifications')
    .upsert({
      yacht_id: yachtId,
      user_id: userId,
      imo_number: results.inputIMO,
      vessel_name: results.inputName,
      imo_valid: results.imo?.found || false,
      imo_matches_name: results.imo?.name?.toLowerCase() === results.inputName?.t
      ais_found: results.ais?.found || false,
      ais_last_position: results.ais?.lat ? {
        lat: results.ais.lat,
        lng: results.ais.lng,
        timestamp: results.ais.lastSeen,
        port: results.ais.lastPort,
      } : null,
      ais_current_status: results.ais?.status || 'unknown',
      equasis_found: results.safety?.found || false,
      equasis_deficiencies: results.safety?.deficiencies || 0,
      equasis_detentions: results.safety?.detentions || 0,
      verification_score: results.score,
      overall_status: results.status,
      verified_at: new Date().toISOString(),
      expires_at: expiresAt.toISOString(),
      raw_response: results,
    }, { onConflict: 'yacht_id' });

  return { data, error };
}
```

ENVIRONMENT VARIABLES (add to .env)

```
VESSELFINDER_API_KEY=your_key_here
```

```
EQUASIS_USERNAME=your_username_here  
EQUASIS_PASSWORD=your_password_here
```

Store in Render environment variables (not in code).

DISPLAY IN MY YACHT PROFILE

After verification, show badge on yacht card:

Aurelia	[Edit]	
Azimut · 24m · 2019		
✓ VERIFIED	Score 94/100	← green badge
Valid until Aug 2026		
[🇮🇹 Valuations]	[📅 17 Charters]	
[🛡️ Verified]	[🔑 Passport]	

TOOLS TAB UPDATE

Update the Yacht Verification card:

```
{  
  id: 'yacht_verification',  
  icon: 'shield-check',  
  title: 'Yacht Verification',  
  subtitle: 'IMO · AIS · Flag registry · Safety records',  
  roles: ['Owner', 'Broker', 'Surveyor'],  
  status: 'LIVE',  
}
```

FILE STRUCTURE

```
/screens/YachtVerification/  
  YachtVerificationScreen.js ← entry + yacht selector  
  VerificationInputScreen.js ← manual input if needed
```

VerificationRunningScreen.js	← progress animation
VerificationResultScreen.js	← results + badges
VerificationDetailScreen.js	← per-check detail view
/components/YachtVerification/	
VerificationBadge.js	← reusable badge component
CheckResultCard.js	← per-check result card
ScoreGauge.js	← circular score display
VerificationCertificate.js	← PDF certificate builder
/utils/	
verificationAPI.js	← API calls (VesselFinder + Equasis)
verificationScore.js	← score calculation
verificationCertPDF.js	← certificate PDF generation

PRIORITY BUILD ORDER

1. verificationAPI.js — VesselFinder + Equasis API calls
 2. YachtVerificationScreen.js — entry + yacht selector
 3. VerificationRunningScreen.js — progress with live updates
 4. VerificationResultScreen.js — main results display
 5. CheckResultCard.js — individual check cards
 6. ScoreGauge.js — score display
 7. Save to Supabase
 8. Badge on My Yacht card
 9. Certificate PDF generation
 10. Add to Digital Passport (future)
-

IMPORTANT NOTES

1. API keys stored as environment variables on Render — NEVER in client code
2. All API calls made from Render backend (Node.js server) — NOT from mobile app directly

3. Mobile app calls YOUR backend endpoint → backend calls VesselFinder/Equasis
4. This protects API keys and avoids CORS issues
5. Verification result valid for 90 days — show expiry date prominently
6. If IMO not provided → skip IMO check, run name-based AIS search only
7. Equasis is FREE but requires registration at equasis.org before integrating
8. VesselFinder Professional plan: €100/month, 5,000 API calls
9. Show clear disclaimer: automated check only, not a substitute for professional survey
10. If API call fails → show "Could not complete check" for that specific item, not full failure

*YachtWorth — Yacht Verification APIs: VesselFinder (€100/mo) + Equasis (free) Checks:
IMO · AIS · Flag Registry · Safety Records · Maritime Liens Certificate valid 90 days · Score
0–100*